

Meridian Parts

European service-hub expansion

Executive consulting report | Synthetic client | Analysis of 2025 shipped sales and returns known through 31 January 2026

Board recommendation	Resource envelope	Base annual increment	Defined stress	Simple payback
CZE+ESP	€225,000 / 5 FTE	€198,099	€32,180	1.14 years

Decision statement

Recommend a staged launch in Czechia, Spain. It ranks first among feasible alternatives on base incremental contribution after recurring fixed cost (€198,099), remains positive under the defined joint stress (€32,180), and uses €225,000 capex/5 FTE. The conclusion is a planning recommendation, not a causal or statistically proven demand forecast.

The recommendation assumes the board wants to stay inside the explicit €450,000 capex, seven-FTE and two-hub constraints, screen out options that are negative under the defined joint stress, and then maximize the base annual contribution. That risk screen is a stated planning choice, not a measured board preference. NLD+ESP is the stronger stress-resilience alternative; its lower base contribution and higher resource requirement are the main concessions of selecting it.

Evidence status

Core operating data are synthetic client exports from the frozen source room. Population, GDP per capita and ECB rates are archived official snapshots preserved with their original URLs, retrieval vintage and hashes. No live client account, customer interview or live public refresh was used. Figures are estimates under the supplied scenario policy, not causal or statistically proven demand forecasts.

1. Executive recommendation

Fund a staged CZE+ESP expansion envelope, releasing Czechia first and making the Spain release conditional on the 90-day gate. In the policy's steady-state arithmetic, the pair produces the highest feasible base incremental contribution: €198.1k per year after recurring fixed cost, with €225k year-zero capex, five FTE and 1.14-year simple payback. The low-volume case remains positive at €64.6k; the defined joint stress remains positive at €32.2k.

Why this option

- 2025 contribution is strongest in ESP (€421,900) and CZE (€397,335); the pair also has the highest base annual scenario value.
- CZE+ESP is positive under the policy's low, base, high and stress cases. The defined stress includes a refund shock for both hubs and an additional 10% net-sales depreciation shock for CZE only.
- The strongest base alternative is POL+ESP at €186,329; it needs €240k/6 FTE and is positive in stress but gives up base value and one additional FTE versus CZE+ESP.
- The choice is reversible after the pilot: defer the second release if realized incremental volume fails the low-case gate or if resource quotes breach the policy limits.

Board decision frame

Alternative	Base annual increment	Stress	Capex / FTE	Comment
CZE+ESP	€198,099	€32,180	€225,000 / 5	Recommended
NLD+ESP	€166,931	€107,630	€270,000 / 6	Stress-resilience alternative
POL+ESP	€186,329	€29,264	€240,000 / 6	Dominated by CZE+ESP on four scenarios and resources
CZE	€95,383	€-37,827	€95,000 / 2	Comparator
DEFER	€0	€0	€0 / 0	Comparator

The board should treat the table as a planning comparison: capex is year-zero, while annual increments exclude capex and use the policy's simple uplift/savings arithmetic. First-year ramp timing is not modelled in payback.

2. Evidence, controls and accounting

The analysis reconciles both order extracts, correction rows, returns, effective-dated unit costs, archived FX history and archived World Bank responses. The source room is frozen; its files were downloaded and hashed into the project before analysis.

Control	Observed	Handling
Orders	1,328 raw rows / 1,297 unique IDs	Highest numeric revision wins; 31 duplicate rows removed; 18 corrections replace full prior rows.
Eligibility	1,254 eligible orders	24 cancelled, 18 test and 1 2026 shipment excluded. Valid zero-price shipments retained (12).
Returns	264 raw rows / 243 IDs	Highest revision wins; repeated IDs counted once; 241 returns linked to eligible orders.
Return cutoff	Inclusive through 2026-01-31	One post-cutoff record excluded; orphan R-ORPHAN quarantined; no guess-based joins.
Missing required values	0 in eligible orders; 0 in selected returns	No imputation needed. Effective unit cost found for every eligible order.
Reconciliation	Monthly ↔ country totals: exact to cent	All monthly records sum to country/year totals; JSON and workbook carry the same values.

Revenue, cash and contribution are different lenses

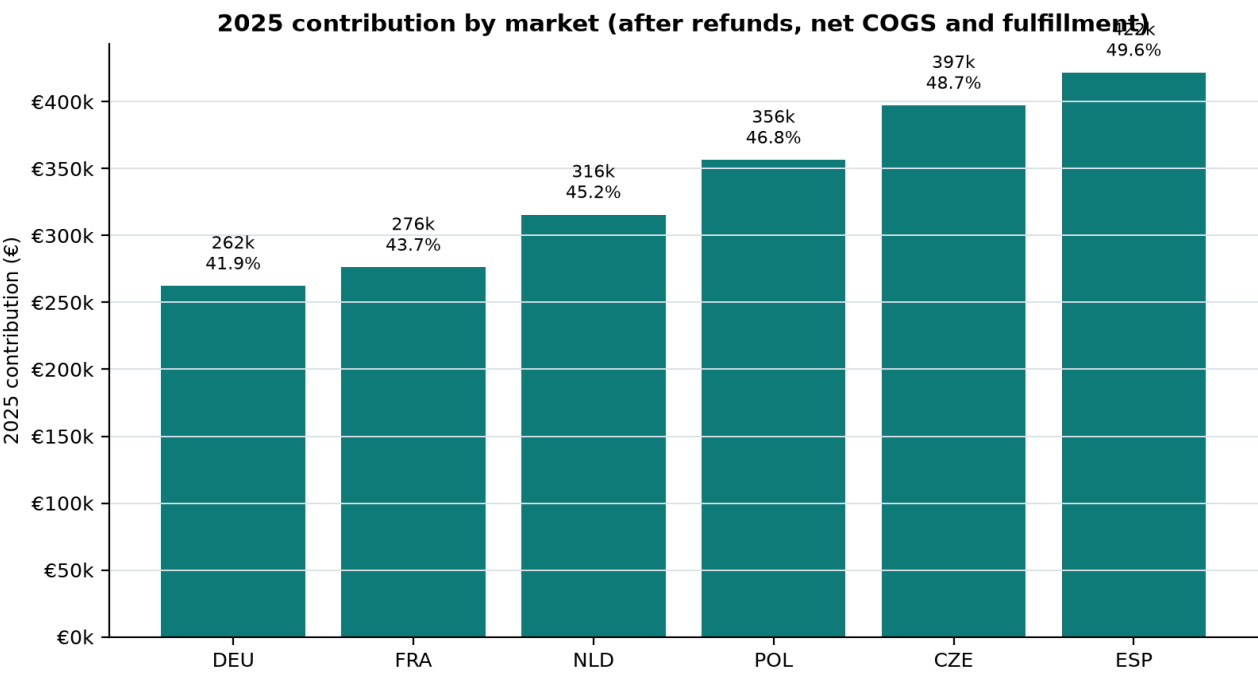
Gross sales are the shipment-date booked sales measure after local-price discounts and original sale-month FX translation. Net sales subtract explicit refunds known by the cutoff; refund receipt dates can lag the shipment month. Cash cannot be reconciled because payment settlement, tax, working-capital and payment-fee data are absent. Contribution is net sales less net COGS and actual nonrefundable fulfillment cost; it is not cash, EBITDA or free cash flow and excludes hub capex by design.

The order detail is the audit trail: gross sales and refunds are converted with the sale-month quote; gross COGS, recovered COGS, net COGS, fulfillment and contribution are rounded per order using half-up cents before sums. Returned units are physical quantity, not number of return records.

3. 2025 operating diagnosis

All six markets have 209 eligible shipped orders after controls. ESP and CZE combine the largest shipped-unit bases with the highest contribution; the result is useful for hub sizing but does not prove that local hubs caused or would cause the uplift in the scenarios.

Country	Orders	Units	Gross sales	Refunds	Net sales	Net COGS	Fulfillment	Contribution	Mar
DEU	209	12,153	€631,245	€5,453	€625,792	€349,980	€13,446	€262,366	41.9
FRA	209	11,880	€640,930	€8,699	€632,231	€341,453	€14,507	€276,271	43.7
NLD	209	12,576	€709,109	€11,376	€697,733	€365,496	€16,593	€315,644	45.2
POL	209	13,163	€776,508	€14,618	€761,890	€386,698	€18,698	€356,494	46.8
CZE	209	13,733	€834,572	€19,265	€815,307	€397,062	€20,910	€397,335	48.7
ESP	209	13,931	€872,259	€21,420	€850,839	€406,230	€22,708	€421,900	49.6

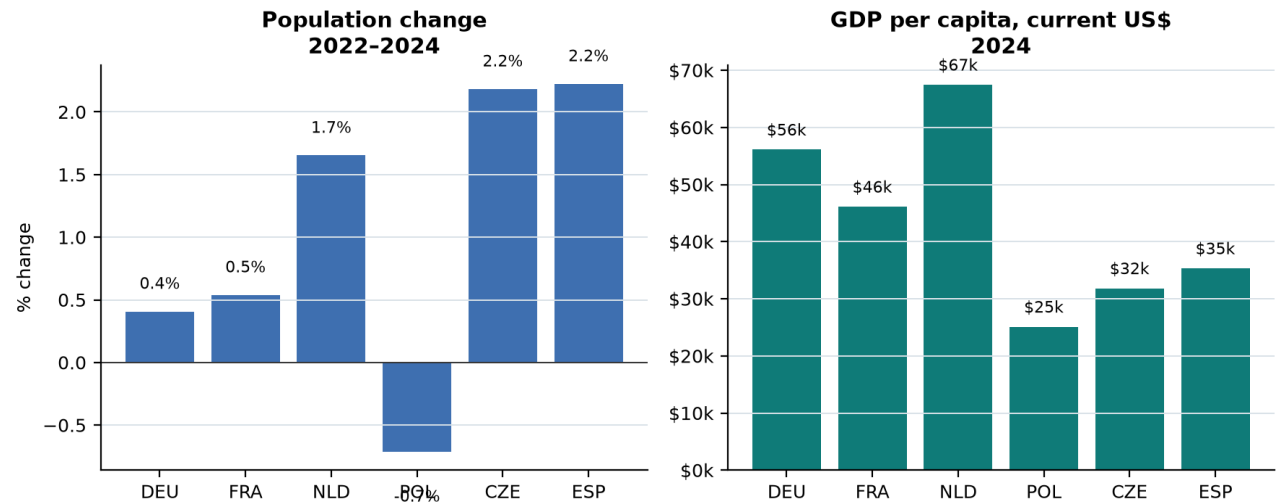


Diagnosis: the operating base is profitable in every market on the stated contribution measure, but the contribution ranking is partly a function of the synthetic order mix, prices, returns and cost assumptions. ESP's 49.6% margin and CZE's 48.7% are not market benchmarks; they are computed from the client extract under the data dictionary.

4. Market context and FX discipline

Archived World Bank context shows different scale and income profiles: NLD and DEU have the highest 2024 GDP per capita in current US\$, while ESP, NLD, DEU, FRA and CZE grow in population between 2022 and 2024 and POL declines. These observations may inform capacity and labour-market conversations, but neither population nor GDP per capita is direct evidence of replacement-assembly demand.

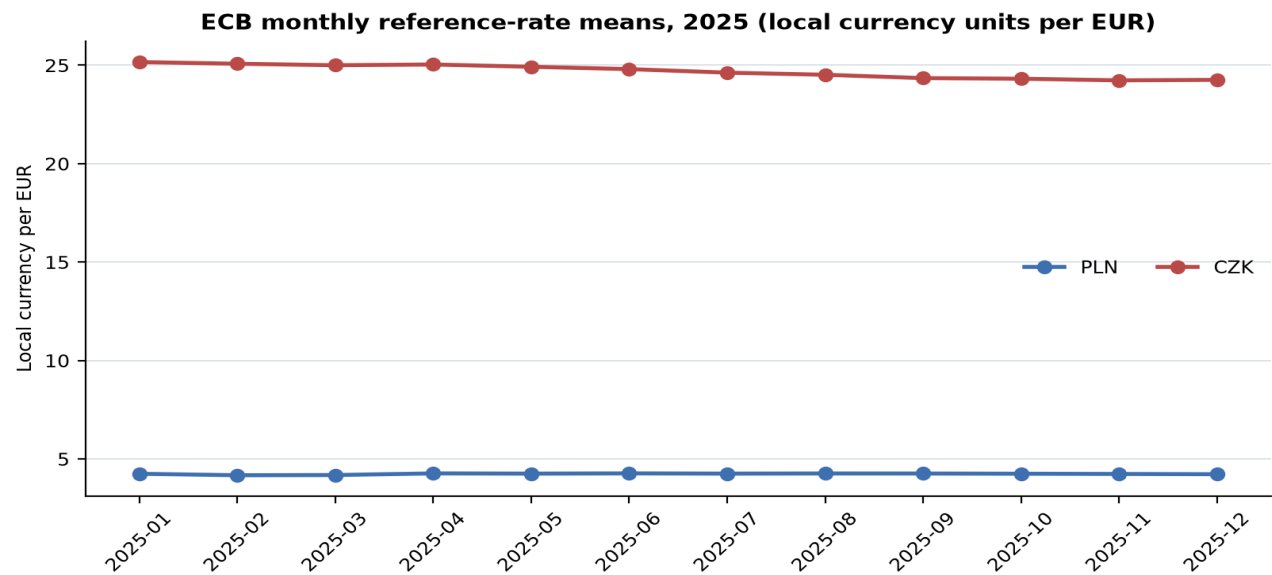
Market context is context, not direct demand evidence



Country	Population 2022	Population 2024	Change	GDP pc 2022	GDP pc 2024
DEU	83,177,813	83,516,593	0.4%	\$50,507	\$56,104
FRA	68,184,457	68,551,653	0.5%	\$40,989	\$46,103
NLD	17,700,982	17,993,485	1.7%	\$59,123	\$67,465
POL	36,821,749	36,559,233	-0.7%	\$18,891	\$25,104
CZE	10,672,118	10,905,028	2.2%	\$28,282	\$31,828
ESP	47,786,102	48,848,840	2.2%	\$30,319	\$35,327

Historical FX is not today's FX

The local-currency orders in POL and CZE use the original sale month's arithmetic mean of available ECB business-day reference observations, quoted as local currency units per EUR. No quote is inverted and no current rate is substituted. The twelve monthly values are in the workbook and JSON; the full archived ECB file and its ZIP hash are in the source register.

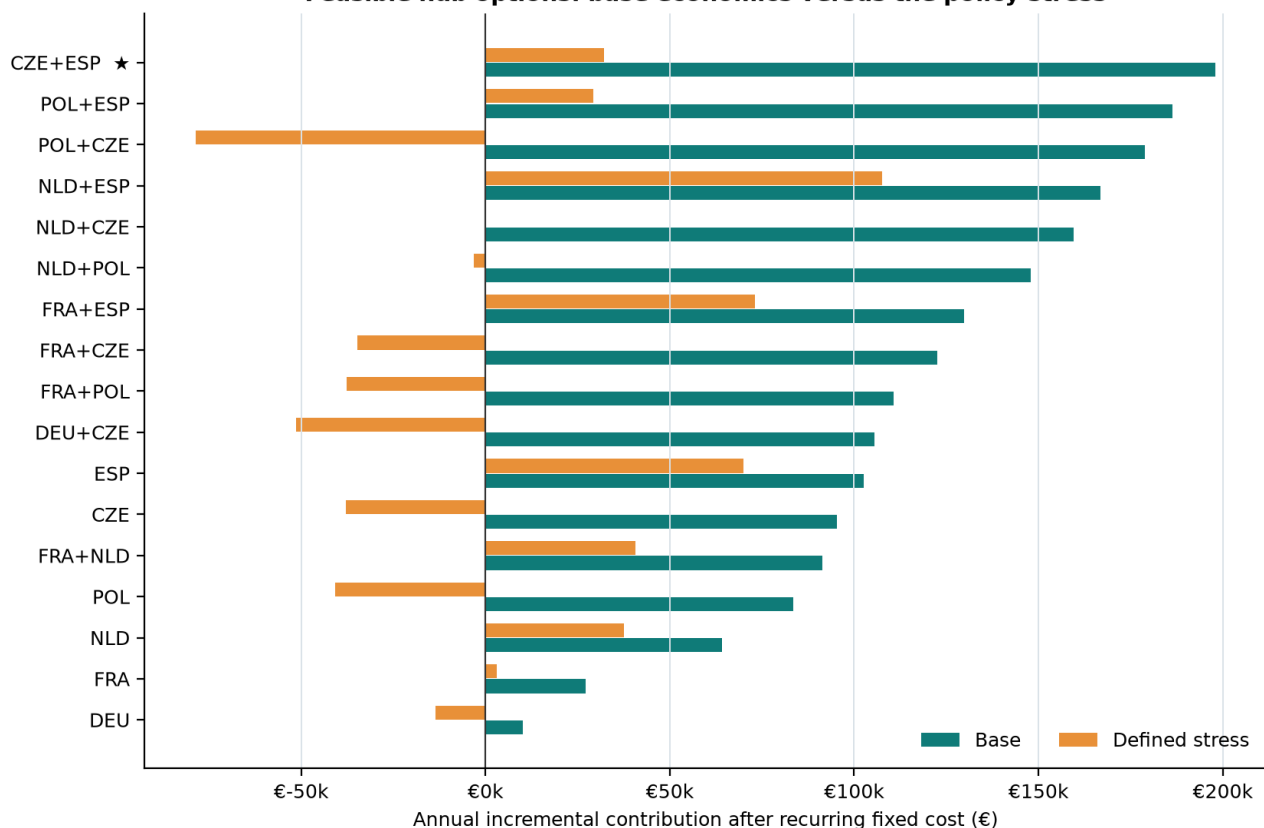


5. Hub economics and scenario comparison

The policy formula is explicit: annual incremental contribution = $C \times \text{uplift} + U \times (1 + \text{uplift}) \times \text{saving per unit} - \text{recurring fixed cost}$. Capex is year-zero and is not subtracted from that annual measure. Simple payback is capex divided by positive annual increment; it is null when annual increment is non-positive. Pairs add country figures without synergy.

Option	Low	Base	High	Stress	Capex	FTE	Base payback
CZE+ESP	€64,619	€198,099	€331,580	€32,180	€225,000	5	1.14y
POL+ESP	€58,562	€186,329	€314,096	€29,264	€240,000	6	1.29y
POL+CZE	€56,856	€178,995	€301,134	€-78,570	€205,000	5	1.15y
NLD+ESP	€43,994	€166,931	€289,869	€107,630	€270,000	6	1.62y
NLD+CZE	€42,289	€159,598	€276,907	€-203	€235,000	5	1.47y
NLD+POL	€36,232	€147,828	€259,423	€-3,119	€250,000	6	1.69y
FRA+ESP	€13,687	€129,849	€246,012	€73,105	€340,000	7	2.62y
FRA+CZE	€11,981	€122,516	€233,050	€-34,729	€305,000	6	2.49y
FRA+POL	€5,924	€110,745	€215,566	€-37,645	€320,000	7	2.89y
DEU+CZE	€-2,549	€105,472	€213,492	€-51,410	€375,000	7	3.56y

Feasible hub options: base economics versus the policy stress



CZE+ESP is the base-case leader and remains positive in the policy stress. NLD+ESP is the stress leader (€107.6k) but produces €31.2k less base annual contribution, consumes €45k more capex and one additional FTE. POL+ESP is dominated by CZE+ESP on base, low, high and stress values while also requiring more capex and FTE; it is not the strongest alternative once those criteria are held constant.

6. Trade-offs, assumptions and what would change the decision

Hard constraints are the client policy's €450k maximum capex, seven FTE maximum and at most two hubs; defer is allowed. The comparison then uses an assumed planning screen: keep positive annual contribution in the defined stress, then maximize base annual contribution. This makes the risk tolerance visible instead of hiding it in a blended score.

Criterion	Status / direction	Evidence or provenance
Capex / FTE / hub count	Hard constraints	Scenario-policy.md; client-supplied synthetic policy.
Base annual increment	Computed; higher is better	Reproducible formula using 2025 country totals and hub-options.csv.
Stress remains positive	Assumed non-compensable screen	Board scenario-policy.md; stress is a conservative joint assumption, not probability.
Payback	Computed; shorter is better, secondary	Simple undiscounted K / positive annual increment; no ramp or discounting.
Market context	Public context only	Archived World Bank responses; not a demand forecast or weight.

Key switching conditions

Condition	Implication
Board prioritizes stress resilience over base value	NLD+ESP becomes the preferred alternative: highest defined-stress contribution (€107.6k), but higher capex/FTE and lower base value.
Normal volume uplift falls below approximately 2.7% for CZE+ESP	The normal annual formula falls below break-even before stress; do not release the second hub without observed demand evidence.
Actual quotes exceed €225k or five FTE for CZE+ESP	Reprice the option and re-screen against policy caps; defer or select a lower-resource alternative.
Observed pilot uplift is below the 10% low case for two consecutive months	Hold the Spain release and reassess; the scenario arithmetic is not a causal guarantee.
Refund/FX stress drivers occur together	Do not treat the positive stress output as certainty; extend pilot, preserve cash and use the gate.

The first-year realization will be lower than the steady-state annual number if the hubs ramp during the 90-day plan. The model does not discount timing, include hiring lag, or model inventory duplication; that limitation is material to payback and is why the release is staged.

7. Staged 90-day implementation

Approve the CZE+ESP envelope but stage execution: prepare both sites, launch CZE first, and release ESP only at Gate 2. The plan translates the model's flip conditions into observable operating evidence.

Timing	Owner	Actions / dependencies	Decision gate
Days 0–15 Mobilize	COO sponsor; Finance; PMO	Confirm baseline definitions and country/order mapping; open capex and labour quotes; appoint CZE/ESP leads; confirm WMS/returns data feeds. Dependency: source-room reconciliation and policy limits.	Gate 0: approve pilot charter; no irreversible lease or hiring commitment until quotes and data ownership are signed off.
Days 16–30 Design	Operations; Procurement; HR; IT	Shortlist local service/3PL sites; map inventory and service levels; verify staffing plan for five FTE; design cross-border returns and restock controls. Dependency: vendor bids, legal/tax review and inventory policy.	Gate 1: CZE and ESP quotes within €225k/5 FTE envelope; service design passes operational risk review.
Days 31–60 Build and test	Country leads; IT/WMS; Quality	Recruit/train, configure routing and stock locations, test order allocation, returns, FX and contribution reporting; run parallel dry-run from central hub. Dependency: clean event timestamps and acceptance test data.	Gate 2: CZE launch only after ≥95% test-order accuracy and no unresolved returns/control defect; reserve Spain release.
Days 61–90 Pilot and release	COO; CZE lead; Spain lead; Finance	Soft-launch CZE; review weekly uplift, OTIF, returns and realized per-unit savings; prepare Spain release package. Dependency: 30 days of reliable CZE KPI data.	Gate 3: release Spain only if annualized CZE volume signal is at least the 10% low case, OTIF ≥95%, returns not >3 pp above baseline, and quotes remain within envelope; otherwise defer.

KPI plan

KPI	Definition / cadence	Gate target / action
Incremental volume uplift	(eligible shipped units in hub market ÷ 2025 baseline run-rate) – 1; weekly and monthly.	≥10% low-case signal by Gate 3; below for two months = hold Spain.
OTIF / service time	Hub-routed orders delivered on-time/in-full; weekly.	≥95% pilot target; below triggers process correction before expansion.
Returns	Returned units ÷ shipped units, by country and SKU; weekly.	No more than 3 percentage points above baseline at Gate 3; investigate root cause.
Realized savings	(central-hub benchmark fulfillment cost – hub fulfillment cost) ÷ shipped units; monthly.	Track against €2.1/unit CZE and €3.0/unit ESP assumptions; reprice if missed.
Contribution after fixed cost	Net sales – net COGS – fulfillment – recurring hub cost; monthly close.	Must be positive in base tracking; Finance owns reconciliation.
Data/control quality	Order, return and FX completeness; monthly close.	Zero orphan joins; all revisions and cutoff flags reconciled.

8. Risks, limitations and evidence register

Risk	Evidence / consequence	Mitigation
Scenario uplift is unmeasured	The 10/25/40% uplift is a synthetic assumption; the decision could overstate hub contribution.	Stage CZE first, use the KPI gate, and do not treat the scenarios as causal estimates.
FX/refund joint stress	Policy stress subtracts 10% of net sales for PLN/CZK and 3% of gross sales refunds; CZE is exposed.	Use monthly sale-month FX translation, monitor net sales/refunds, and hold Spain if stress drivers emerge.
Capex, fixed cost and FTE estimates	Hub options are planning inputs, not vendor quotes or labour offers.	Obtain signed bids and staffing quotes before Gate 1; re-screen hard caps.
First-year ramp / inventory duplication	Simple payback is steady-state and undiscounted; timing and working capital are absent.	Stage releases and add a first-year cash/stock model before lease signature.
Data quality drift	A future order, orphan return and duplicate/revised rows were present in the source room.	Retain the revision/cutoff controls as a monthly close process; do not guess orphan mappings.

Source register summary

ID	Saved file	Original URL / status	Period / units
S1	source-room-index.html	http://127.0.0.1:55463/source_room_index	collection-time source index HTML source inventory
S2	data-dictionary.md	http://127.0.0.1:55463/data-dictionary.md synthetic_client	2025; cutoff 2026-01-31 2025 shipments/returns; policy metadata
S3	scenario-policy.md	http://127.0.0.1:55463/scenario-policy.md synthetic_client	2025 baseline and annual scenarios EUR, units, annual contribution, capex, FTE
S4	hub-options.csv	http://127.0.0.1:55463/hub-options.csv synthetic_client	annual planning assumptions EUR capex/fixed cost/savings; FTE; country
S5	unit-costs.csv	http://127.0.0.1:55463/unit-costs.csv synthetic_client	effective 2025-01-01 and 2025-07-01 EUR per unit
S6	orders-part1.csv	http://127.0.0.1:55463/orders-part1.csv synthetic_client	2025 and excluded test/future rows local currency prices; EUR fulfillment; units
S7	orders-part2.csv	http://127.0.0.1:55463/orders-part2.csv synthetic_client	2025 and excluded test/future rows local currency prices; EUR fulfillment; units
S8	order-corrections.csv	http://127.0.0.1:55463/order-corrections.csv synthetic_client	higher revisions replacing order rows local currency prices; EUR fulfillment; units
S9	returns.csv	http://127.0.0.1:55463/returns.csv synthetic_client	2025-01-28 to 2026-02-01 local currency refunds; physical units
S10	population.json	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000 official_public_archive	2022-2024 persons
S11	gdp-per-capita.json	https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000 official_public_archive	2022-2024 current US\$ per person
S12	ecb-history.zip	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip official_public_archive	2025 observations used; file contains broader history local currency units per EUR
S13	ecb-history.csv	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip official_public_archive	2025 observations used; file contains broader history local currency units per EUR

The full source register, hashes, archived retrieval metadata, saved raw files, analysis script, workbook, metrics JSON and board deck are delivered with this report. Source room index: <http://127.0.0.1:55463/>. Original public URLs are retained in deliverables/source-register.json; the local source room is a frozen evidence room, not a live account.

Appendix. Reproducibility and formulas

- Order eligibility: status = shipped, is_test = false, and shipped_at in calendar year 2025 after selecting the highest numeric revision per order_id.
- Gross sales EUR: $\text{half_up}((\text{quantity} \times \text{unit_price_local} - \text{discount_local}) \div \text{sale-month local_per_eur}, 2)$. Refunds EUR: $\text{half_up}(\text{sum explicit refund_local on usable returns} \div \text{sale-month local_per_eur}, 2)$.
- Net COGS EUR: $\text{half_up}(\text{half_up}(\text{quantity} \times \text{effective unit cost}, 2) - \text{half_up}(\text{restocked units} \times \text{effective unit cost}, 2), 2)$. Contribution = gross sales – refunds – net COGS – fulfillment, all order-level components half-up rounded to cents.
- Margin = contribution $\div$ net sales; null when net sales is zero. Monthly and country tables use the same order-level rows, so reconciliation is direct.
- Scenario low/base/high = $C \times u + U \times (1+u) \times s - F$ for $u = 0.10/0.25/0.40$. Stress = $(C - 0.03 \times G - \text{FX_shock}) \times 1.25 - C + U \times 1.25 \times s - F$, where $\text{FX_shock} = 0.10 \times N$ for PLN/CZK markets and zero otherwise.
- A reviewer can reproduce all figures by running ``python analysis/reproduce.py`` from the project root. The script reads only evidence/raw and writes derived outputs under deliverables/.

Completion note

The deliverables complete the requested analytical exchange. The recommendation remains conditional because demand uplift, hub savings, fixed costs, vendor quotes and first-year ramp were supplied as assumptions rather than observed causal evidence. The stated gates and flip conditions are therefore part of the recommendation, not footnotes.